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Lifestyle interventions for hypertension management in primary care: a narrative review

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Hypertension is one of the leading chronic diseases globally and a major contributor to cardiovascular morbidity and mortality. Despite advances in pharmacological therapy, medication alone remains limited in achieving optimal control. This review synthesizes recent hypertension management guidelines, including those from the European Society of Cardiology (ESC, 2024), American Heart Association/American College of Cardiology (AHA/ACC, 2025), Taiwan Society of Cardiology/Hypertension Society (2022), and Korean Society of Hypertension (KSH, 2018). All guidelines consistently emphasize sodium restriction, weight reduction, regular exercise, moderation of alcohol intake, smoking cessation, and adoption of healthy dietary patterns such as the Dietary Approaches to Stop Hypertension, Mediterranean, or culturally adapted diets. The ESC 2024 guideline elevates lifestyle modification to Class I, Level A, specifying targets for sodium (<2 g/day) and potassium (≥3.5 g/day). The AHA/ACC 2025 guideline provides quantitative estimates, reporting approximately 1/1 mm Hg blood pressure reduction per kilogram of weight loss, and incorporates newer strategies such as glucagon-like peptide-1 receptor agonists and bariatric surgery when lifestyle measures alone are insufficient. Taiwan's 2022 guideline frames recommendations under the S-ABCDE (sodium restriction, alcohol limitation, body weight reduction, cigarette cessation, diet adaptation, exercise adoption) mnemonic and uniquely includes genetic factors such as *ALDH2* polymorphisms. The KSH 2018 guideline emphasizes salt restriction (<6 g/day), maintaining a body mass index <25 kg/m^2, and adherence to traditional Korean diets. Lifestyle modification remains the cornerstone of hypertension prevention and management, particularly in primary care. Future directions should focus on integrating these approaches with pharmacotherapy, digital health strategies, and personalized prescriptions.

Keywords: Hypertension; Life style; Diet; DASH; Sodium; Guideline

Introduction

Hypertension is among the most prevalent chronic diseases worldwide and a leading cause of cardiovascular morbidity and mortality [1]. The current definition of hypertension is based on the blood pressure (BP) threshold at which pharmacological treatment has been shown to be effective in randomized clinical trials. Accordingly, the Korean Society of Hypertension (KSH) defines hypertension as a systolic blood pressure (SBP) ≥140 mm Hg or a diastolic blood pressure (DBP) ≥90 mm Hg, further classified into grade 1 and grade 2 hypertension based on severity and office BP measurements [2]. Normal BP is defined as SBP <120 mm Hg and DBP <80 mm Hg, which is associated with the lowest cardiovascular risk and serves as the reference category when evaluating the risks of elevated BP values [3,4].

In Korea, the burden of hypertension continues to rise. According to the KSH fact sheet 2023, the prevalence of hypertension among adults aged ≥20 years is approximately 28%, representing nearly 12.3 million individuals. Among those with hypertension, awareness was 74.1%, treatment 70.3%, and control 56.0% [5].

Despite advances in pharmacological therapy, antihypertensive medication alone is insufficient for optimal BP control and long-term complication prevention. Consequently, lifestyle modification has been consistently emphasized as the cornerstone of both prevention and management of hypertension [2,5-10]. The KSH guidelines explicitly recommend lifestyle modification alongside pharmacotherapy, with the 2018 update highlighting that interventions such as salt restriction and weight loss can lower BP by amounts comparable to a single antihypertensive drug [2]. International guidelines similarly emphasize that lifestyle modification

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should continue even when medications are prescribed. Among various interventions, the Dietary Approaches to Stop Hypertension (DASH) diet has demonstrated robust effects, reducing SBP by about 11 mm Hg and DBP by 6 mm Hg in controlled studies—comparable to first-line antihypertensive agents [11]. Korean observational evidence supports these findings. In a primary care cohort study, reduced salt intake and increased physical activity were significantly associated with improved BP control, whereas failure to implement these behaviors nearly doubled the likelihood of uncontrolled hypertension [12]. Consistent with global evidence, additional studies in Korean populations confirm that reducing sedentary behavior and engaging in regular physical activity are crucial for lowering BP [13,14]. These findings underscore the clinical importance of incorporating lifestyle interventions into routine hypertension management, particularly in primary care settings where continuous patient education and monitoring are feasible [15].

Recently updated guidelines from the KSH [3], the American Heart Association/American College of Cardiology (AHA/ACC) [9], and the European Society of Cardiology (ESC) [8] have further refined lifestyle management recommendations. This review summarizes these updated recommendations, focusing on their applicability in primary care practice.

Methods

Ethics statement

As this study is based entirely on published literature, neither approval from an institutional review board nor the obtainment of informed consent was required.

Study design

This narrative review examined 4 major hypertension guidelines—those from the United States (ACC/AHA 2025), Europe (ESC 2024), Korea (KSH 2018), and Taiwan (TSOC/THS 2022). These were selected based on 4 key criteria: recency, accessibility, applicability to Korean primary care, and inclusion of Asian-specific factors. The Taiwanese guideline was specifically chosen because it represents the most recent and English-accessible Asian guideline that explicitly incorporates genetic and cultural factors—most notably the *ALDH2* polymorphism ("Asian flush")—into its recommendations on alcohol consumption. This population-specific element is particularly relevant to Koreans, who share similar genetic and lifestyle backgrounds. By contrast, the Japanese (JSH 2019) and Chinese (2024 revision) guidelines were excluded from the main comparative synthesis for practical and methodological reasons.

Results

The 2024 European Society of Cardiology guideline

The 2024 ESC guideline positions lifestyle modification as a central strategy in hypertension management and provides the following specific recommendations [8].

Dietary sodium and potassium

For adults with elevated BP or hypertension, sodium intake should be restricted to approximately 2 g/day (equivalent to 5 g of salt or roughly 1 small teaspoon) (Class I, Level A) [16]. In patients without moderate-to-advanced chronic kidney disease (CKD) and with high sodium consumption, increasing potassium by 0.5–1.0 g/day through potassium-enriched salt (75% NaCl/25% KCl) or diets rich in fruits and vegetables is advised (Class IIa, Level A) [17,18]. For those with CKD or those receiving potassium-sparing medications (e.g., certain diuretics, angiotensin converting enzyme inhibitors, angiotensin receptor blockers, or spironolactone), serum potassium should be monitored when increasing dietary potassium (Class IIa, Level C) [8]. The World Health Organization recommends a daily potassium intake of at least 3.5 g; excessive supplementation should be avoided, and potassium limited to <2.4 g/day in advanced CKD [19].

Weight reduction and diet

A stable, healthy body mass index (BMI; 20–25 kg/m^2) and waist circumference below 94 cm for men and 80 cm for women are recommended (Class I, Level A). On average, a 5 kg weight loss reduces SBP/DBP by approximately 4.4/3.6 mm Hg, and a 13% reduction in BMI from 40 kg/m^2 is associated with a 22% lower risk of incident hypertension. Sustained 5%–10% weight loss improves BP, glucose regulation, and lipid metabolism, and may reduce premature mortality [20,21]. Glucagon-like peptide-1 (GLP-1) receptor agonists, such as semaglutide, can reduce both body weight and BP [22]. Adoption of a Mediterranean or DASH diet is strongly recommended (Class I, Level A), with additive BP-lowering effects when combined with weight loss and sodium restriction [23].

Physical activity and exercise

Regular aerobic exercise can reduce SBP by up to 7–8 mm Hg and DBP by up to 4–5 mm Hg. In non-White individuals with hypertension, dynamic resistance training may produce BP reductions comparable to those achieved with aerobic activity. Recommended targets include at least 150 minutes per week of moderate-intensity aerobic exercise (≥30 minutes on 5–7 days per week) or 75 minutes per week of vigorous-intensity exercise spread over

at least 3 days. These should be complemented by low- to moderate-intensity dynamic or isometric resistance training 2 to 3 times per week (Class I, Level A) [24]. High-intensity interval training produces BP reductions comparable to moderate continuous exercise and may yield superior fitness gains [25]. However, high-intensity exercise should be avoided in individuals with uncontrolled resting hypertension.

Alcohol, coffee, and sugar-sweetened beverages

Alcohol intake should be limited to ≤100 g of pure alcohol per week, with complete abstinence encouraged for optimal cardiovascular outcomes (Class I, Level B) [26]. Coffee consumption is not associated with increased hypertension risk and may even confer protective effects; evidence for tea is mixed but suggests potential BP-lowering benefits [27]. In contrast, energy drinks can elevate BP and increase cardiovascular complications. Free sugar intake should be restricted to ≤10% of total daily energy [28], and the consumption of sugar-sweetened beverages should be discouraged from early life (Class I, Level B).

Smoking

Complete abstinence from tobacco is strongly recommended (Class I, Level A), with proactive counseling and referral to cessation services [29]. The effects of e-cigarettes on BP remain uncertain, though emerging evidence indicates a potential for BP elevation [30].

The 2025 AHA/ACC guideline

The 2025 AHA/ACC guideline identifies lifestyle modification as a core strategy for all adults with hypertension and those at elevated risk [9].

Sodium restriction

Lower sodium intake reduces BP across the life span and helps prevent incident hypertension. Compared with an intake of approximately 4,500 mg/day, a low-sodium diet (≤1,500–2,300 mg/day) reduces BP by about 3/2 mm Hg in normotensive individuals and by 7/3 mm Hg in those with hypertension [31,32]. Because most dietary sodium derives from processed and restaurant foods, population-level interventions are needed to achieve sustained reduction [33].

Salt substitutes and potassium

Potassium-enriched salt formulations (e.g., 75% NaCl/25% KCl) reduce BP by approximately 3.3–5/1.5 mm Hg and, in large clinical trials, decrease the incidence of stroke, major adverse cardiovascular events, and all-cause mortality by about 12%–14%, without increasing hyperkalemia risk [17]. Increased dietary potassium from fruits, vegetables, and legumes—or moderate supplementation—lowers BP by about 6/4 mm Hg, particularly in individuals with hypertension and high sodium intake. However, supplementation exceeding 80 mmol/day should be avoided, and caution is advised for patients with CKD or those taking potassium-retaining drugs [34].

Weight reduction

For non-Asian adults with BMI 25.0–29.9 kg/m^2 or ≥30 kg/m^2 and Asian adults with BMI 23.0–27.4 kg/m^2 or ≥27.5 kg/m^2, weight loss is a key intervention for improving overall health and controlling BP [35]. On average, each kilogram of weight loss reduces BP by approximately 1/1 mm Hg (SBP/DBP) [20]. A reduction of ≥5% of body weight or ≥3 kg/m^2 in BMI yields larger BP decreases, regardless of hypertension status, and shows synergistic effects when combined with the DASH diet and sodium restriction [36]. When lifestyle interventions alone are insufficient, pharmacologic options (e.g., GLP-1 receptor agonists) or bariatric surgery may be considered [37].

Dietary pattern—DASH

The DASH dietary pattern remains the most evidence-based approach for BP control. It emphasizes fruits, vegetables, low-fat dairy products, and whole grains, supplying potassium, magnesium, calcium, and fiber [36]. Clinical trials demonstrate SBP reductions of 1–13 mm Hg and DBP reductions of 1–10 mm Hg, with greater effects observed among Black adults, individuals with higher baseline BP, and those with high sodium consumption [38]. BP-lowering effects are amplified when DASH is combined with weight reduction or sodium restriction.

Alcohol

Any baseline alcohol consumption is associated with increased long-term BP. The risk of incident hypertension rises progressively with greater alcohol intake, with abstainers showing the lowest risk. Reducing alcohol intake by ≥50%, particularly in individuals consuming ≥4–6 drinks per day, leads to meaningful BP reductions, whereas no significant BP improvement is seen when reducing intake from ≤2 drinks per day [39]. Given the balance of risks, abstinence appears optimal [40].

Physical activity

Leisure-time physical activity lowers BP in adults with hypertension [41]. Aerobic exercise reduces BP by approximately 4–7/3–4 mm Hg, dynamic resistance training by 3/2 mm Hg,

and isometric training by 8/4 mm Hg. A dose–response relationship exists, with an additional 2/1 mm Hg BP reduction per extra 30 minutes per week of aerobic activity, reaching maximal benefit at about 150 minutes per week [42]. Both continuous and interval training formats are effective.

Stress reduction

Short-term studies show that transcendental meditation can reduce BP by approximately 5/2 mm Hg, paced breathing by 5/3 mm Hg, and yoga by smaller but consistent amounts [43]. These interventions serve as supportive measures, although their effects are generally less pronounced than those achieved through diet, sodium restriction, or structured exercise [44].

In summary, the AHA/ACC 2025 guideline underscores weight reduction, the DASH diet, sodium restriction, potassium optimization, alcohol limitation (preferably abstinence), regular exercise, and stress management as central lifestyle strategies capable of achieving BP reductions comparable to those of pharmacologic therapy.

2022 Guidelines of the Taiwan Society of Cardiology and the Taiwan Hypertension Society for the Management of Hypertension

The Taiwan guideline summarizes lifestyle therapy using the S-ABCDE framework: sodium restriction, alcohol limitation, body weight reduction, cigarette cessation, diet adaptation, and exercise adoption. Lifestyle modification is central to both prevention and management of hypertension; however, sustaining long-term adherence remains a key challenge. Therefore, the guideline emphasizes cognitive-behavioral and multimodal strategies to enhance persistence with these interventions [10].

Sodium restriction

A daily sodium intake of 2–4 g (equivalent to 5–10 g of salt) is recommended. Taiwanese and international evidence links high sodium intake with increased stroke incidence and cardiovascular mortality. The Salt Substitute and Stroke Study demonstrated that potassium-enriched salt reduced SBP and stroke incidence by 14% [17]. Because a J-curve relationship has been observed at intakes below 2 g/day, the guideline advocates realistic and sustainable sodium restriction rather than overly aggressive targets [45].

Alcohol limitation

Abstainers should not be encouraged to start drinking. For those who consume alcohol, intake should be limited to <100 g/wk for men and <50 g/wk for women. Given the high prevalence (approximately 40%–50%) of the *ALDH2**2 polymorphism in Taiwan, which predisposes to alcohol flushing, stricter limits are recommended: <64 g/wk for men and <28 g/wk for women [46].

Weight reduction

A BMI target of 20–24.9 kg/m^2 is recommended, with a 5 kg weight loss associated with an approximate 4.4 mm Hg reduction in SBP [20]. For individuals with severe obesity, anti-obesity pharmacotherapy or bariatric surgery may be considered.

Smoking cessation

Complete cessation of both conventional and electronic cigarettes is advised. The principal benefit of smoking cessation lies in reducing overall cardiovascular risk rather than producing a direct, sustained BP-lowering effect.

Diet adaptation

The DASH diet is prioritized, with an average BP reduction of about 11/6 mm Hg. While the Mediterranean diet is acceptable, the guideline generally favors DASH, reflecting Taiwan's high stroke burden and genetic context. Both green and black tea may offer modest BP-lowering effects of approximately 1–2 mm Hg [47,48].

Exercise adoption

At least 30 minutes of moderate aerobic activity is recommended on 5–7 days per week, supplemented by resistance or neuromotor exercises such as tai chi, yoga, or meditation [49]. Randomized controlled trials show reductions of approximately 3.5/2.5 mm Hg with aerobic exercise and 10.9/6.2 mm Hg with isometric exercise. Even under levels of PM2.5 (particulate matter with a diameter of 2.5 μm or less) exposure typical in many urban areas, the benefits of regular physical activity outweigh potential risks, and exercise remains strongly encouraged [50].

Overall, the Taiwan guideline positions lifestyle therapy as a foundational pillar of hypertension care—one tailored to Asian genetic, dietary, and environmental contexts—and highlights multimodal, adherence-enhancing strategies as essential for long-term success.

The KSH 2018 guideline

The KSH emphasized the central role of lifestyle modification in both its 2013 and 2018 guidelines [2,6]. Both editions identify

non-pharmacologic measures—healthy diet, regular exercise, smoking cessation, and moderation of alcohol intake—as the foundation of hypertension management, noting that their BP-lowering effects are comparable to those of a single antihypertensive medication (Class I, Level A). The 2018 update advances the 2013 framework by specifying recommendation classes and evidence levels, integrating contemporary domestic data, and improving clinical applicability. Notably, while the 2013 guideline used general descriptors such as "strongly recommended," the 2018 version formally assigns classes and levels of Evidence (e.g., Class I, Level A). It also broadens the target population to include individuals with elevated BP (prehypertension), thereby strengthening the emphasis on prevention.

Sodium restriction

The 2013 guideline reported an average daily salt intake of approximately 12.2 g among Koreans and recommended reducing it to <6 g/day, while acknowledging uncertainty—such as a possible J-curve—regarding the relationship between salt restriction and cardiovascular outcomes. The 2018 guideline incorporates updated Korea National Health and Nutrition Examination Survey data, citing a mean intake of 10 g/day [51], and again recommends <6 g/day (Class I, Level A). Beyond BP reduction, it highlights central hemodynamic considerations for cardiovascular event prevention. The guideline further specifies that salt restriction yields greater benefits in salt-sensitive groups (older adults, individuals with obesity, diabetes, or a family history of hypertension). It adds practical strategies such as increasing low-salt food options, avoiding processed foods high in sodium, and favoring natural ingredients during cooking. Importantly, it newly underscores potassium-rich foods as a means of offsetting sodium-related BP increases, with appropriate caution for patients with CKD or risk of hyperkalemia [3].

Weight management

The 2013 guideline noted that even a 5 kg weight loss can reduce BP and described associations among abdominal obesity, hypertension, metabolic disorders, and cardiovascular mortality. The 2018 update retains these points and incorporates Korean cohort data showing the lowest all-cause mortality at BMI 23.0–24.9 kg/m^2. It recommends maintaining BMI <25 kg/m^2 and specifies Korean-specific waist circumference thresholds (<90 cm in men and <85 cm in women) [52], offering a broader and more practical approach than the previous 80 cm threshold for women in the 2013 guideline (Class I, Level A).

Alcohol

The 2013 guideline advised 20–30 g/day for men and 10–20 g/day for women, cautioning against the elevated stroke risk in heavy drinkers. The 2018 guideline simplifies this to fewer than 2 drinks per day (Class I, Level A) [53].

Exercise

Whereas the 2013 guideline primarily emphasized aerobic exercise with isometric activity as a supplement, the 2018 update provides detailed prescriptions: 5–7 sessions per week, 30–60 minutes per session (≥90–150 minutes per week total), including 5-minute warm-up and cool-down periods. It also specifies resistance protocols—dynamic resistance (2–3 sets) and isometric handgrip training at 30%–40% of maximal strength, held for 2 minutes with 1-minute rests, repeated 4 times, 3 sessions per week—allowing "recipe-level" guidance for clinical counseling [3] (Class I, Level A).

Smoking cessation

As in 2013, the 2018 guideline mandates complete cessation and supports combined nicotine replacement and behavioral therapy. The update elevates this recommendation to Class I, Level A, underscoring active physician counseling and referral to cessation programs [54] (Class I, Level A).

Dietary management

The 2018 guideline continues to endorse healthy dietary patterns such as the DASH and Mediterranean diets, while emphasizing that a Korean-style diet—rich in tofu, legumes, vegetables, fruits, and fish—is associated with a lower prevalence of hypertension [55,56]. It further reiterates that higher intake of calcium, magnesium, and potassium can yield an additional 11/6 mm Hg BP reduction in hypertensive patients [11,57]. Among older adults, increasing fruit and vegetable intake alone lowers BP by 3/1 mm Hg, with an additional 6/3 mm Hg reduction when combined with reduced fat consumption [11,57].

In summary, the 2018 KSH guideline retains the core principles of 2013 while introducing formalized grading, expanded domestic evidence, increased emphasis on potassium intake, detailed exercise prescriptions, and updated waist circumference thresholds. These refinements enhance practical applicability in real-world care. Lifestyle modification remains a first-line strategy with BP-lowering effects comparable to pharmacotherapy, while offering broad cardiometabolic protection.

Table 1 provides a comparative summary of lifestyle intervention recommendations across recent hypertension guidelines.

Table 1. Comparison of lifestyle modification recommendations for hypertension management across guidelines

Lifestyle factor	Europe (ESC 2024)	USA (AHA/ACC 2025)	Korea (KSH 2018)
Sodium restriction	~2 g sodium/day (≈5 g salt); supported by long-term outcome trials (SSaSS, DECIDE-Salt)	1,500–2,300 mg/day (optimal <1,500); salt substitutes reduce SBP by ~3.3 mm Hg	<6 g salt/day (≈2.4 g sodium)
Potassium intake	≥3.5 g/day (WHO target); increased intake recommended unless advanced CKD	Diet rich in fruits/vegetables; moderate supplementation lowers BP by ~6/4 mm Hg; avoid >80 mmol/day	Potassium-rich foods encouraged (esp. with salt restriction); caution in CKD
Weight reduction	Stable BMI 20–25 kg/m^2; WC <94 cm (men), <80 cm (women); 5 kg loss → SBP ↓ 4.4 mm Hg	≥5% body weight or ≥3 kg/m^2 BMI loss; ~1/1 mm Hg per kg loss; GLP-1 RAs or bariatric surgery if needed	Target BMI <25 kg/m^2; lowest mortality at 23–24.9 kg/m^2; WC <90 cm (men), <85 cm (women)
Dietary pattern	Mediterranean or DASH diet strongly recommended	DASH diet most effective (SBP ↓ 1–13, DBP ↓ 1–10 mm Hg); Mediterranean, vegetarian also beneficial	DASH or Korean-style diet (vegetables, tofu, fish, fruits) associated with lower HTN prevalence
Alcohol consumption	<100 g/wk pure alcohol; ideally abstain	Abstinence optimal; <2 drinks/day men, <1 drink/day women; dose-response ↑ risk	<20–30 g/day men, <10–20 g/day women
Exercise	≥150 min/wk moderate or 75 min vigorous aerobic; + resistance 2–3×/wk; avoid high intensity if uncontrolled HTN	Aerobic: 150 min/wk (SBP ↓ 4–7 mm Hg); dynamic resistance: ↓ 3/2; isometric: ↓ 8/4	≥30–50 min/day, ≥5 days/wk aerobic; isometric exercise 2–3×/wk; caution with isotonic heavy lifting
Smoking cessation	Strongly recommended; link to CV mortality; e-cigarettes not recommended	Strongly recommended; part of CV risk reduction	Strongly recommended at every visit; nicotine replacement + behavioral therapy
Sugar intake	Restrict free sugar <10% of daily energy; discourage soft drinks/fruit juices early in life	Restrict free sugar <10% of daily energy; avoid sugar-sweetened beverages	Not specifically emphasized in 2018 KSH
Stress management	Stress reduction (mindfulness, relaxation) considered beneficial, though evidence limited	Meditation ↓ ~5/2 mm Hg; breathing ↓ 5/3 mm Hg; yoga modest effect	Stress reduction acknowledged but evidence insufficient

ESC, European Society of Cardiology; AHA, American Heart Association; ACC, American College of Cardiology; KSH, Korean Society of Hypertension; SSaSS, Salt Substitute and Stroke Study; DECIDE-Salt, Diet, Exercise and Cardiovascular Health–Salt Trial; SBP, systolic blood pressure; WHO, World Health Organization; CKD, chronic kidney disease; BP, blood pressure; BMI, body mass index; WC, waist circumference; GLP-1 RA, glucagon-like peptide-1 receptor agonist; DASH, Dietary Approaches to Stop Hypertension; DBP, diastolic blood pressure; HTN, hypertension; CV, cardiovascular.

Discussion

This review compared and analyzed lifestyle modification strategies across the most recent hypertension guidelines from Europe, the United States, Taiwan, and Korea. All 4 guidelines consistently emphasize sodium restriction, weight reduction, regular exercise, smoking cessation, moderation of alcohol intake, and adherence to healthy dietary patterns such as the DASH or Mediterranean diets as the foundation of hypertension management. Collectively, these interventions provide BP-lowering effects comparable to those achieved with a single antihypertensive medication.

The ESC 2024 guideline elevates lifestyle modification to a Class I recommendation, equivalent in importance to pharmacologic therapy, and establishes specific intake targets for sodium (<2 g/day) and potassium (≥3.5 g/day). It underscores the broader health benefits of lifestyle interventions beyond BP control, highlighting their roles in improving metabolic health and reducing overall cardiovascular risk. Furthermore, the guideline calls for coordinated societal and policy-level measures to strengthen adherence and expand population-level impact.

The AHA/ACC 2025 guideline provides detailed quantitative evidence, reporting an approximate 1/1 mm Hg BP reduction per kilogram of weight loss and up to 13/10 mm Hg reduction with adherence to the DASH diet, and places particular emphasis on integrating adjunctive strategies. These include the use of anti-obesity pharmacotherapy, such as GLP-1 receptor agonists, and bariatric surgery when lifestyle measures alone fail to achieve adequate BP control.

The Taiwan Society of Cardiology/Hypertension Society (2022) guideline organizes its recommendations using the mnemonic S-ABCDE (sodium restriction, alcohol limitation, body weight reduction, cigarette cessation, diet adaptation, exercise adoption). It uniquely incorporates genetic considerations, particularly *ALDH2* polymorphisms associated with "Asian flush," into its alcohol recommendations, and integrates culturally relevant components such as tea consumption, meditation, and tai chi into its lifestyle framework.

The KSH (2018) guideline reflects the local context of high-

salt dietary patterns and the substantial burden of cerebrovascular disease in Korea. It strongly recommends reducing salt intake to <6 g/day and promotes Korean-style dietary patterns rich in tofu, legumes, fish, vegetables, and fruits. Specific anthropometric targets include maintaining BMI <25 kg/m^2 and waist circumference <90 cm for men and <85 cm for women. The 2022 Focused Update of the KSH guidelines primarily addressed pharmacologic treatment strategies, diagnostic thresholds, and management of special populations. Notably, it did not revise lifestyle modification recommendations. Therefore, the principles of non-pharmacologic management, including sodium restriction, weight control, regular physical activity, moderation of alcohol intake, smoking cessation, and adherence to dietary patterns such as the DASH diet, remain those established in the 2018 guideline. In current clinical practice, these 2018 recommendations continue to serve as the standard reference for lifestyle-based hypertension management in Korea.

Mechanisms of blood pressure reduction by lifestyle modification

Lifestyle modification lowers BP through multiple interrelated physiological pathways. Sodium restriction decreases extracellular fluid volume and cardiac output, thereby reducing systemic BP [58]. Weight reduction improves insulin sensitivity, attenuates sympathetic nervous system activity, and diminishes renal sodium reabsorption [20]. Regular physical activity enhances endothelial function and arterial compliance, leading to decreased peripheral vascular resistance [59]. Limiting alcohol intake reduces sympathetic stimulation and suppresses activation of the renin–angiotensin–aldosterone system [53]. Increased potassium consumption promotes natriuresis and induces relaxation of vascular smooth muscle, contributing to vasodilation [60]. Lastly, stress management mitigates hyperactivation of the sympathetic nervous system and the hypothalamic–pituitary–adrenal axis, further supporting BP regulation [44].

Strengths and limitations of this review

The strength of this review lies in its comprehensive synthesis of the most recent national and international hypertension guidelines, supported by evidence from large-scale meta-analyses. By comparing recommendations from Korea, the United States, Europe, and Taiwan, this review provides practical, contextually relevant strategies for implementing lifestyle modification in primary care settings. Additionally, the quantitative estimation of BP-lowering effects across individual interventions offers clinicians clinically meaningful guidance for patient management.

Nevertheless, several limitations should be acknowledged. First, the magnitude of BP reduction achieved through lifestyle modification varies across studies, depending on the intensity, duration, and adherence to interventions. Second, much of the available evidence is derived from Western populations, whereas long-term, prospective data from Korean and other Asian cohorts remain limited. Third, although short- and medium-term BP reductions are well documented, the long-term effects of lifestyle interventions on cardiovascular morbidity and mortality are still insufficiently established.

Conclusion

In conclusion, all major hypertension guidelines reaffirm lifestyle modification as the cornerstone of hypertension prevention and management. Core components—including sodium restriction, weight reduction, regular physical activity, a balanced diet, smoking cessation, and moderation of alcohol intake—are universally endorsed. However, regional distinctions exist. The United States emphasizes quantitative evidence and integration of obesity therapeutics; Europe underscores policy-level and population-based strategies; Korea focuses on culturally adapted low-salt diets; and Taiwan incorporates genetic and cultural specificity, particularly in its alcohol-related recommendations. Building on these perspectives, the Korean guideline could be further enhanced by defining explicit quantitative targets for sodium and potassium intake (e.g., Na <2 g/day, K ≥3.5 g/day) and by integrating Asian-specific genetic considerations such as the *ALDH2* polymorphism influencing alcohol metabolism. Moreover, greater attention to sleep hygiene, stress regulation, and mental health would support a more holistic approach to hypertension care within primary practice. Looking forward, the incorporation of digital health technologies, artificial intelligence-driven precision approaches, and genetic or microbiome-informed personalization will be pivotal to optimizing lifestyle-based hypertension management, particularly among Asian populations, including Koreans.

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Conceptualization: BH, YHK. Data curation: GBL, JY. Methodology: BH, YHK. Investigation: BH, YHK. Writing–original draft: BH, YHK. Writing–review & editing: BH, GBL, JY, YHK.

Conflict of interest

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Data availability

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Supplementary materials

None.

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